

KESHAV BHANDARI

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SUMMARY

I am a PhD student at the Centre for Digital Music (C4DM) at Queen Mary University of London. My research revolves around neuro-symbolic approaches for generating music with a focus on controllability and musical structure.

EDUCATION

Queen Mary University of London, School of Electronic Engineering and Computer Science <i>PhD in Computer Science – Neuro Symbolic Automated Music Composition</i>	Sep 2023 – Present London, UK
Northwestern University, McCormick School of Engineering <i>Master of Science in Artificial Intelligence</i>	Sep 2021 – Dec 2022 Evanston, IL, USA
Purdue University, Krannert School of Management <i>Bachelor of Science in Business Management & Marketing (Dean's List, 2013)</i>	Aug 2012 – May 2016 West Lafayette, IN

PUBLICATIONS

- **Bhandari, K.**, Chang, S., Lu, T., Enus, F. R., Herremans, D., & Colton, S. (2025). ImprovNet - Generating Controllable Musical Improvisations with Iterative Corruption Refinement. In International Joint Conference on Neural Networks (IJCNN).
- **Bhandari, K.**, Wiggins, G. A., & Colton, S. (2025). Yin-Yang: Developing Motifs With Long-Term Structure And Controllability. In International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART).
- **Bhandari, K.**, Roy, A., Wang, K., Puri, G., Colton, S., & Herremans, D. (2025). Text2midi: Generating Symbolic Music from Captions. Proceedings of the AAAI Conference on Artificial Intelligence (AAAI).
- Colton, S., Bradshaw, L., Banar, B., & **Bhandari, K.** (2024). Automatic Generation of Expressive Piano Miniatures. International Conference on Computational Creativity (ICCC).
- **Bhandari, K.**, & Colton, S. (2024). Motifs, Phrases, and Beyond: The Modelling of Structure in Symbolic Music Generation. In International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART).
- O'Reilly, P., Bugler, A., **Bhandari, K.**, Morrison, M., & Pardo, B. (2022). VoiceBlock: Privacy through Real-Time Adversarial Attacks with Audio-to-Audio Models. Advances in Neural Information Processing Systems (NeurIPS).

GRANTS AWARDED

- “Using deep learning to build a robust automated recognition system for rare and endangered birds in India” **Oct 2021**
PI: **Keshav Bhandari**
AI for Earth Microsoft Azure Research Credits Grant. **\$5,000**.
Role: I substantially wrote and edited this grant, and it was based on my work.

PAPER AWARDS

- EvoMUSART (Part of EvoSTAR) Best Paper Award, 2025:
Keshav Bhandari, Geraint A. Wiggins, Simon Colton. Yin-Yang: Developing Motifs With Long-Term Structure And Controllability. In the 2025 International Conference on Artificial Intelligence in Music, Sound, Art and Design.
Documentation: <https://www.seresearch.qmul.ac.uk/cmai/news/4970/cmai-best-paper-award-at-evomusart-2025/>

GUEST LECTURES

- Queen Mary University of London, School of Electronic Engineering and Computer Science. Class: Google Deepmind Research Ready Programme Orientation. Professor: Simon Colton. June 26th, 2025. Talk Title: Limitations of Sequential Models and An Introduction To Transformers.
- Queen Mary University of London, School of Electronic Engineering and Computer Science. Class: Computational Creativity. Professor: Simon Colton. April 2nd, 2025. Talk Title: Self-Supervised Corruption-Refinement Models for Low Resource Generative Music.
- Queen Mary University of London, School of Electronic Engineering and Computer Science. Class: Google Deepmind Research Ready Programme Orientation. Professor: Simon Colton. June 28th, 2024. Talk Title: How To Be An Expert At Literature Reviews.

INVITED TALKS

- Northwestern University, McCormick School of Engineering. Group: Interactive Audio Lab. Professor: Bryan Pardo. Feb. 13th, 2025. Talk Title: ImprovNet - Generating Controllable Musical Improvisations with Iterative Corruption Refinement.

RESEARCH PRESENTATIONS

- Queen Mary University of London, Digital Music Research Network One-Day Workshop (DMRN+18). December 2023. Talk Title: Towards Melodic Development with Discrete Diffusion Models for Symbolic Music.

- Digital University Kerala, 11th International Conference on Ecological Informatics (ICEI 2020+1). November 2021. Talk Title: Augmentations to improve rare bird call classification for a highly imbalanced multi-label soundscape environment.

TEACHING

- Demonstrator for ECS7022P Computational Creativity *Jan 2024 - Present*
- Senior Demonstrator for Data Science and AI master's cohort *Sep 2024 - Present*
- Demonstrator for ECS7024P - Statistics for Artificial Intelligence and Data Science *Sep 2024 – Dec 2024*
- Teaching Fellow – Supervising 6 UG and 12 PG students on their final year projects *Oct 2024 - Present*
- Google DeepMind Research Ready Programme - PhD Mentor for two students on a research project *Jun 2024 – Jul 2024*

CONFERENCE SERVICE

- Session Chair, Special Track: Human-AI Interaction in Creative Arts and Sciences, International Joint Conference on Neural Networks (IJCNN), Rome, Italy – July 3, 2025.
- Reviewer, International Joint Conference on Neural Networks (IJCNN), 2025.

NEWSPAPER ARTICLES & MENTIONS

- Saving every tweet: Meet researchers using bird calls, AI to study biodiversity. Hindustan Times. <https://bit.ly/43rVJLv>

PROFESSIONAL EXPERIENCE

Boston Consulting Group – Data Scientist (Los Angeles, CA, USA) *Jan 2023 – Sep 2023*

- Vodafone Italy: Developed AI-based tools using Llama and Stable Diffusion to enhance digital marketing efficiency.
- Staples: Designed a pattern matching algorithm to improve control group selection for precise reporting and strategic insights.
- Walgreens: Built model drift scripts to rectify deviations in price optimization algorithm in a MLOps framework.

Epsilon – Senior Data Scientist (Bangalore, Karnataka, India) *Sep 2017 – Feb 2021*

- Built deep learning based personalized recommendation engines using Transformers, boosting client hit rates by up to **5%**.
- Deployed end-to-end ML & DL solutions as a product on cloud platforms, achieving lift rates between **3X to 9X**.
- Developed Gredel, an interactive R-Shiny package for automating predictive modeling and improving ML interpretability.

ACCOMPLISHMENTS

- EvoSTAR 2025 Best Paper Award / Outstanding Student Award *2025*
- 2nd place in Epsilon India Recommendation Engine Hackathon (26 Teams) *2020*
- Top 6% Kaggle Categorical Feature Encoding Challenge (1,342 Teams) *2019*
- Champion Innovator Spot Award, Epsilon *2019*
- Top 3% Kaggle TalkingData Ad-Tracking Fraud Detection Challenge (3,946 Teams) *2018*

LEADERSHIP & COMMUNITY SERVICE

Data Kind, Bangalore Chapter *Aug 2018 – Oct 2019*

President, Purdue Marketing Association, Purdue University *Jan 2015 - Dec 2015*